INCH-POUND
A-A-59653
31 August 2001
SUPERSEDING
FF-B-187/5
10 September 1993

COMMERCIAL ITEM DESCRIPTION

BEARING, ROLLER, TAPERED, DOUBLE ROW, ONE DOUBLE CONE, TWO SINGLE CUPS (TYPES TDI AND TDIS)

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

1. SCOPE. This commercial item description (CID) establishes government acquisition requirements for double row, tapered, roller bearings with one double cone and two single cups (types TDI and TDIS) for general purpose use.

2. CLASSIFICATION. The roller bearings shall be of the following types (see 7.4(b)) and sizes:

Type: Type I - TDI (normal angle)
Type II - TDIS (steep angle)

Size: See table I (Column headings in table I refer to characteristics defined in figure 1.)

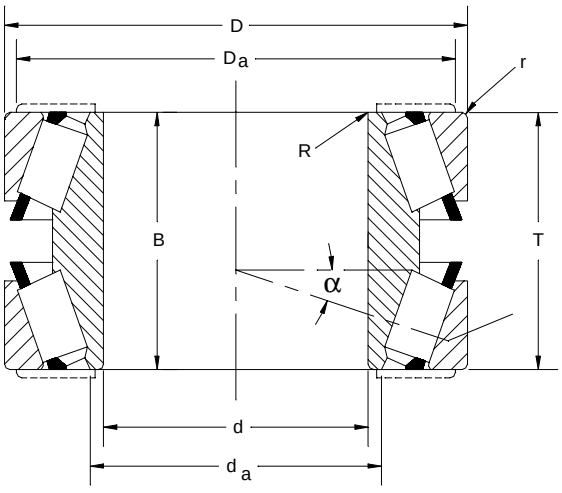


FIGURE 1. Bearing configuration.

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data that may improve this document should be sent to: Defense Supply Center Richmond (DSCR), ATTN: DSCR-VBD, 8000 Jefferson Davis Highway, Richmond, VA 23297-5610.

AMSC N/A FSC 3110

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TABLE I. Size codes and dimensions.*

Size code	Part 1/ number	d	D	T	B	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
001	17116D-17244	1.1875	2.4409	1.3306	1.3750	0.03	0.06	1.40	2.13	1.53	12000
002	14126D-14276	1.2500	2.7170	1.5632	1.5426	0.06	0.05	1.57	2.36	1.53	15100
003	19145D-19268	1.4375	2.6875	1.4492	1.5000	0.03	0.06	1.67	2.40	1.31	14200
004	19145D-19281	1.4375	2.8125	1.4492	1.5000	0.03	0.04	1.667	2.48	1.31	14200
005	19145D-19283	1.4375	2.8346	1.5392	1.5000	0.03	0.06	1.67	2.48	1.31	14200
006	358D-354A	1.6875	3.3465	1.9790	2.0620	0.06	0.05	2.01	3.03	1.91	22800
007	358D-352	1.6875	3.5480	2.1650	2.0620	0.06	0.09	2.01	3.07	1.91	22800
008	376DE-374	1.8750	3.6718	1.9690	2.0940	0.03	0.05	2.17	3.35	1.37	24400
009	375D-374	2.0000	3.6718	1.9690	2.0940	0.03	0.05	2.24	3.35	1.73	24400
010	399D-394A	2.4375	4.3307	2.1870	2.1870	0.03	0.05	2.76	3.98	1.45	27700
011	399D-394AS	2.4375	4.3307	2.1870	2.1870	0.03	0.13	2.76	3.90	1.45	27700
012	39585D-39520	2.5000	4.4375	2.3750	2.3750	0.03	0.13	2.83	3.98	1.72	46200
013	39585D-39521	2.5000	4.4375	2.3750	2.3750	0.03	0.03	2.83	4.06	1.72	46200
014	78251D-78537	2.5000	5.3750	2.6020	2.5980	0.09	0.13	3.10	4.53	0.67	55200
015	78251D-78551	2.5000	5.5130	2.6020	2.5980	0.09	0.09	3.11	4.61	0.67	55200
016	LM114848D-LM114811	3.0000	4.5625	2.1250	2.1250	0.03	0.06	3.27	4.21	2.15	34600
017	765D-752	3.0000	6.3750	3.9980	4.2500	0.14	0.13	3.75	5.75	1.71	89900
018	496D-492A	3.1875	5.2500	2.3750	2.3440	0.06	0.13	3.58	4.72	1.31	42800
019	496D-493	3.1875	5.3750	2.3750	2.3440	0.06	0.13	3.58	4.80	1.31	42800

1/ Part numbers are for reference only. Part numbers reflect the cup and cone numbers used by industry and the American Bearing Manufacturers Association (ABMA).
2/ These maximum fillet radii shall be cleared by the bearing corners.
* Column headings in the above table refer to bearing characteristics defined in figure 1.

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	B	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
020	581D-572	3.1875	5.5115	3.1875	3.1549	0.06	0.13	3.62	4.92	1.45	51100
021	L116149DA-L116110	3.2500	4.5625	1.6250	1.9375	0.06	0.06	3.54	4.25	1.90	15000
022	L217845D-L217810	3.3750	4.8750	1.6251	1.7500	0.03	0.06	3.66	4.57	1.77	25900
023	L217849DA-L217810	3.5000	4.8750	1.6251	1.9063	0.06	0.06	3.82	4.57	1.77	25900
024	767D-752A	3.5000	6.3750	3.9980	4.2500	0.06	0.13	3.99	5.75	1.71	89700
025	865D-854	3.5000	7.5000	4.6250	5.0000	0.38	0.13	4.57	6.69	1.74	125400
026	42362D-42584	3.6250	5.8437	2.2500	2.2812	0.06	0.12	4.06	5.28	1.19	45300
027	LM119348D-LM119311	3.7500	5.3750	2.2500	2.2500	0.03	0.09	4.02	4.96	2.06	38700
028	868D-854	4.0000	7.5000	4.6250	5.0000	0.06	0.13	4.57	6.69	1.74	125400
029	945D-932	4.0000	8.3750	5.6250	6.0000	0.41	0.13	5.25	7.36	1.79	142500
030	LM121948D-LM121911	4.2500	6.0000	2.5625	2.5625	0.03	0.13	5.51	4.53	2.07	49500
031	71426D-71750	4.2500	7.5000	3.8750	4.0000	0.06	0.13	4.80	6.73	1.40	100000
032	946D-932	4.2500	8.3750	5.6250	6.0000	0.13	0.13	5.04	7.36	1.79	142500
033	71450D-71750	4.5000	7.5000	3.8750	4.0000	0.06	0.13	5.04	6.73	1.40	100000
034	HH224346DD-HH224310	4.5000	8.3750	5.6250	6.0000	0.13	0.13	5.24	7.56	1.79	185600
035	LM124449D-LM12441C	4.7500	6.5625	2.8440	2.8440	0.03	0.13	5.04	6.06	1.99	58000
036	M224749D-M224711	4.7500	6.7812	2.6251	2.6875	0.03	0.06	5.08	6.38	1.76	68000

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* Column headings in the above table refer to bearing characteristics defined in Figure 1.

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	B	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
037	M224749D-M224710	4.7500	6.8750	2.6251	2.6875	0.03	0.06	5.08	6.42	1.76	68000
038	48290W-48220	5.0000	7.1875	3.0000	3.0000	0.06	0.13	5.39	6.61	1.91	74000
039	67388D-67322	5.0000	7.7500	3.6250	3.6250	0.06	0.13	5.50	7.09	1.70	97800
040	EE153053D-153100	5.0000	10.0000	6.3750	6.7500	0.13	0.25	5.88	8.86	1.82	205200
041	74510D-74845	5.1183	8.4636	4.8750	4.8750	0.06	0.13	5.75	7.72	1.20	98800
042	74510D-74850	5.1183	8.5000	4.8750	4.8750	0.06	0.13	5.75	7.72	1.20	98800
043	67390D-67322	5.2500	7.7500	3.6250	3.6250	0.06	0.13	5.71	7.09	1.70	97800
044	48393D-48320	5.3750	7.5000	3.0625	3.0625	0.06	0.13	5.67	6.97	1.82	75000
045	H228649D-H228610	5.3750	8.8750	4.7500	4.7500	0.06	0.13	5.98	7.99	1.76	177500
046	48680D-48620	5.5000	7.8750	3.0625	2.9688	0.03	0.13	5.91	7.28	1.74	74600
047	74555D-74845	5.5000	8.4636	3.7500	4.1875	0.06	0.13	6.06	7.72	1.20	99000
048	74555D-74850	5.5000	8.5000	3.7500	4.1875	0.06	0.13	6.06	7.72	1.20	99000
049	M231649D-M231610	6.0000	8.7500	3.3125	3.3125	0.06	0.06	6.50	8.15	1.76	94300
050	99603D-99100	6.0000	10.0000	6.2500	6.2500	0.06	0.13	6.69	8.94	1.43	156800
051	HH234049D-HH234018	6.0000	12.5000	6.3750	6.7500	0.38	0.27	7.15	11.02	1.79	333400
052	450900D-451212	6.0000	12.1250	6.3750	6.7500	0.38	0.27	7.38	10.59	1.79	252300
053	46790D-46720	6.5000	8.8750	3.1250	3.0000	0.03	0.13	6.89	8.23	1.52	78200
054	46791DW-46720	6.5000	8.8750	3.1250	3.1250	0.03	0.13	6.89	8.23	1.52	78200
055	67790D-67720	7.0000	9.7500	3.5625	3.5625	0.06	0.13	7.48	9.02	1.33	66800

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TABLE I. Size codes and dimensions (continued).*

Size code	Part 1/ number	d	D	T	B	R 2/	r 2/	d a	D a	K factor	Basic dynamic load ratings (lb.)
		Bore	Outside diameter	Bearing width over cups	Cone width	Max. shaft fillet radius	Max. housing fillet radius	Recommended shoulder diameter			
	Cone-cup							Shaft	Housing		Two row radial
056	82681D-82620	7.0000	11.0000	4.4374	4.4375	0.06	0.13	7.68	9.88	1.11	90800
057	EE91700D-91112	7.0000	11.2500	4.1874	4.1875	0.06	0.13	7.64	10.28	1.35	134000
058	94706D-91113	7.0000	11.3750	4.8750	4.8750	0.06	0.13	7.68	10.20	1.35	174200
059	HM237546D-HM237510	7.0000	11.3750	4.8750	4.8750	0.06	0.13	7.64	10.47	1.83	211800
060	HM237546DD-HM237510	7.0000	11.3750	6.2500	6.2500	0.06	0.13	7.63	10.47	1.83	211800
061	HM237546D-HM237513	7.0000	11.4163	4.8750	4.8750	0.06	0.12	7.64	10.51	1.83	211800
062	EE280700D-281200	7.0000	12.0000	4.3086	4.5000	0.13	0.13	7.87	10.98	1.62	174800
063	M238849DW-M238810	7.3750	10.6250	4.0000	4.0000	0.06	0.13	7.87	9.84	1.76	151000
064	H239649D-H239610	7.3750	12.5970	6.6250	6.3750	0.13	0.19	8.23	11.54	1.83	303000
065	H239649D-H239612	7.3750	12.6250	6.6250	6.3750	0.13	0.19	8.23	11.54	1.83	303000
066	67885DW-67820	7.5000	10.5000	3.5625	3.5313	0.06	0.13	8.03	9.69	1.22	115900
067	EE420750D-421437	7.5000	14.3720	6.2500	6.0000	0.13	0.13	8.70	12.95	1.45	337000
068	93800D-93125	8.0000	12.5000	4.8750	4.8750	0.06	0.13	8.74	11.26	1.12	200400
069	93801D-93125	8.0000	12.5000	5.2500	5.2500	0.25	0.13	8.93	11.26	1.12	200400
070	EE420800D-421437	8.0000	14.3720	6.2500	6.0000	0.13	0.13	9.06	12.95	1.45	320200
071	EE420800D-421450	8.0000	14.5000	6.2500	6.0000	0.13	0.13	9.06	13.03	1.45	320200

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2/ These maximum fillet radii shall be cleared by the bearing corners.
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3. SALIENT CHARACTERISTICS

3.1 Dimensions. Bearing dimensions (and dynamic load ratings, see 3.4) shall conform to the requirements specified in table I for each of the bearing part numbers. The listed dimensions conform to the requirements specified in American National Standards Institute/American Bearing Manufacturers Association (ANSI/ABMA) Standard 19.2, "Tapered Roller Bearings - Radial Inch Design" (DoD adopted). The bearing size shall be specified in the acquisition order (see 7.4(c)). For any unlisted bearing size codes, the associated dimensional and dynamic load rating requirements should also be specified in the acquisition order.

3.2 Materials.

3.2.1 Cones (inner rings), cups (outer rings), and rollers. The bearing cones, cups, and rollers shall be made of case carburized or through-hardened steel produced in accordance with the American Society for Testing and Materials (ASTM) A 295, "Standard Specification for High-Carbon Anti-Friction Bearing Steel" (DoD adopted), ASTM A 534, "Standard Specification for Carburizing Steels for Anti-Friction Bearings" (DoD adopted), or ASTM A 535, "Standard Specification for Special-Quality Ball and Roller Bearing Steel" (DoD adopted). The steel shall show a fine fracture grain size in accordance with ASTM E 112, "Standard Test Methods for Determining Average Grain Size" (DoD adopted). Material hardness shall be no less than 58 HRC and no more than 64 HRC as defined in ASTM E 18, "Standard Test Methods for Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials" (DoD adopted).

3.2.2 Cage. The bearing cage material shall be impervious to deterioration from any lubricant, preservative, solvent, or other chemical substance expected to contact the bearing during normal use or storage. Similarly, the material shall not cause any chemical deterioration of any other bearing component. The cages shall be made from carbon steel (one piece stamped). Materials shall operate from -65 to 230 °F.

3.3 Tolerance class. The tolerance limits for bearings shall conform to tolerance class 4 as tabulated in ABMA Standard 19.2. Allowable tolerances for bearing components and assembled bearings are listed in tables II through V.

3.4 Dynamic load rating. The bearing dynamic load rating shall conform to the requirements specified in table I for each bearing size code. The listed ratings conform to the requirements specified in ABMA Standard 11, "Load Ratings and Fatigue Life for Roller Bearings".

3.5 Lubrication and preservation. The bearings shall be furnished without lubrication and shall be preserved for packaging in accordance with MIL-DTL-197, "Packaging of Bearings, Antifriction Associated Parts and Subassemblies". If a preservative compound is required for packing or storage, the compound name shall be specified in the acquisition order (see 7.4(d)).

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TABLE II. Cone bore tolerance.

Cone bore (d)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0	3.0000	5	0
3.0000	8.0000	10	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE III. Cup diameter tolerance.

Cup diameter (D)			
Size range		Tolerance	
Over	Incl.	Plus	Minus
0.0000	12.0000	10	0
12.0000	24.0000	20	0

Note: Allowable tolerances are in 0.0001 inch.

TABLE IV. Bearing width tolerance.

Bearing width (T)			
Bore size range		Tolerance	
Over	Incl.	Plus	Minus
0	4.0000	160	0
4.0000	12.0000	280	200

Note: Allowable tolerances are in 0.0001 inch.

TABLE V. Assembled bearing tolerance.

Assembled bearing maximum radial runout		
Cup OD (D)		Tolerance
Over	Incl.	
0	24.0	20

Note: Allowable tolerances are in 0.0001 inch.

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3.6 Contact angle. All bearings except size codes 14 and 15 are normal angle bearings having a contact angle between 10 and 19 degrees. Bearing size codes 14 and 15 are steep angle bearings having a contact angle between 22 and 31 degrees. The contact angle is the angle between the line of action of the roller load and a plane perpendicular to the bearing axis.

4. REGULATORY REQUIREMENTS

4.1 Recovered materials. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

4.2 Unless otherwise indicated in the solicitation and resulting contract, the foreign acquisition restrictions in Section 225.7019 of the Defense Federal Acquisition Regulation Supplement (DFARS) apply to products described by this CID.

5. PRODUCT CONFORMANCE PROVISIONS

5.1 Product conformance. The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial marketplace. The government reserves the right to require proof of such conformance.

5.2 Market acceptability. The product offered must have been previously sold either to the government or on the commercial market.

6. PACKAGING

6.1 Preservation, packing, and marking. For acquisition purposes, the bearings supplied shall be preserved, packed, and marked as specified in the acquisition order (see 7.4(e)).

7. NOTES

7.1 Sources of documents.

7.1.1 FAR and DFARS. Copies of the FAR and DFARS may be obtained from the Superintendent of Documents, P.O. Box 371954, Pittsburgh, PA 15250-7954. Electronic copies of the FAR may be obtained from http://www.arnet.gov/far/. Electronic copies of the DFARS may be obtained from http://www.acq.osd.mil/dp/dars/dfars.html.

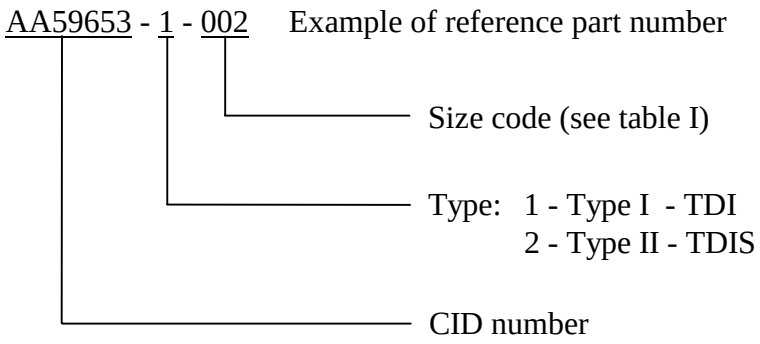
7.1.2 Military details. Copies of military details may be obtained from Standardization Documents Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094. Electronic copies of military specifications and standards may be obtained from http://astimage.daps.dla.mil/quicksearch/.

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7.1.3 ABMA standards. Copies of ABMA standards may be obtained from the American Bearing Manufacturers Association, 1200 19th Street, Suite 300, Washington, DC 20036-2401.

7.1.4 ASTM standards. Copies of ASTM standards may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959.

7.2 Part or identification number (PIN). The following part or identification numbering procedure is for government purposes and does not constitute a requirement for the contractor.



AA59653 - 1 - 002 indicates: Type TDI bearing; bearing bore 1.2500 inches; outside diameter 2.7170 inches; width 1.5632 inches.

7.3 Sources of supply. The manufacturers and/or suppliers listed below are known to supply products that meet the salient characteristics requirements of this document. Competition is not limited to the listed firms.

KOYO Corporation of U.S.A. Westlake, OH 44145	FAG Bearings Corporation Danbury, CT 06813
NTN Bearing Corporation Mount Prospect, IL 60056	SKF Industries Kulpsville, PA 19443

7.4 Ordering data. Acquisition documents shall specify the following information:

a. CID document number, revision, and CID PIN
b. Bearing type (see 2)
c. Bearing size (with dimension/load requirements if size is unlisted) (see 3.1)
d. Preservative compound, if required (see 3.5)
e. Preservation, packing, and marking requirements (see 6.1)

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7.5 Cross-reference information. Table VI relates the original specification slant sheets to the replacement CIDs.

TABLE VI. Cross-reference table.

FF-B-187B	Replacement CID	ABMA type
Specification sheet 1	A-A-59649	TS
Specification sheet 2	A-A-59650	TSF
Specification sheet 3	A-A-59651	TSS
Specification sheet 4	A-A-59652	TSSF
Specification sheet 5	A-A-59653	TDI & TDIS
Specification sheet 6	A-A-59654	TDO
Specification sheet 7	A-A-59655	TDOS
Specification sheet 8	A-A-59656	TNA (normal angle)
Specification sheet 9	A-A-59657	TNAS (steep angle)
Specification sheet 10	A-A-59658	TNASW
Specification sheet 11	A-A-59659	TNASWE

7.6 Subject term (key word) listing.

bore
cone
cup
load
width

MILITARY INTERESTS:

Custodians:
Army - AR
Navy - YD
Air Force - 99

Reviewers:
Navy - MC, OS
Air Force - 84

CIVIL AGENCY
COORDINATING ACTIVITY:

GSA - 7FXE

Preparing activity:
DLA - GS4

(Project 3110-1241)

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